গবেষণা, পরীক্ষণ ও কনসালটেনসি কেন্দ্র
পুর ও পরিবেশ কৌশল বিভাগ
শাহজালাল বিজ্ঞান ও প্রযুক্তি বিশ্ববিদ্যালয়
সিলেট-৩১১৪, বাংলাদেশ।
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Website: www.crtc-cee-sust.com

Center for Research, Testing and Consultancy
Department of Civil and Environmental Engineering
Shahjalal University of Science and Technology
Sylhet-3114, Bangladesh.
Contact: +88 01318 210 286/ 287/ 288/ 289/ 290
Email: crtc-cee@sust.edu

**Ref. no: CRTC/CEE/G20220406/S-63**

# SUB-SOIL INVESTIGATION REPORT

## FOR THE PROPOSED
## SIX STORIED RESIDENTIAL BUILDING

**At Megna B-52, Dariapara, Sylhet**
**DAG # 18561, KHA # 18581, J.L # 91, Mouja- Municipality.**

**CLIENT**

**MOJNU MIAH**
**S/O- CHARAG MIAH**

Verification Code: 9n484U1t
Verify report at: crtc-cee-sust.com

**CONDUCTED BY**

CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET

**SAMPLE COLLECTED BY**

THE UNIVERSAL SOIL & ENGINEERING
MOKTASOR MARKET (2ND FLOOR), SHOP NO-06,
BAGBARI POINT, SYLHET
Mob- 01741217881

## Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would necessary to select proper type of foundation.

## Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

## Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

**Table – 1: Items of the supplied samples**

| No | Name | Bore hole no | Depth | Quantities |
|---|---|---|---|---|
| 01 | Disturbed sample | 01 | 0-55 ft | 11 |
| 02 | Disturbed sample | 02 | 0-55 ft | 11 |
| 03 | Disturbed sample | 03 | 0-55 ft | 11 |
| 04 | Disturbed sample | 04 | 0-60 ft | 12 |
| 05 | Disturbed sample | 05 | 0-60 ft | 12 |
| 06 | Disturbed sample | 06 | 0-60 ft | 12 |
| 07 | - | - | - | - |
| 08 | - | - | - | - |
| 09 | - | - | - | - |
| 10 | - | - | - | - |

## Laboratory Classifications

The disturbed soil samples, supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with bore hole (BH) and depth (Please see enclosed Tables).

## Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

## Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

**Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)**

| N-values | Consistency | Unconfined Compressive Strength ($q_u$) | |
|---|---|---|---|
| | | tsf | kPa |
| 0-2 | Very soft | 0 to 0.25 | 0 to 25 |
| 3-5 | Soft | 0.25 to 0.50 | 25 to 50 |
| 6-9 | Medium | 0.50 to 1 | 50 to 100 |
| 10-16 | Stiff | 1 to 2 | 100 to 200 |
| 17-30 | Very stiff | 2 to 4 | 200 to 400 |
| >30 | Hard | >4 | >400 |

**Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)**

| N values | Compactness | Relative Density, Dr % | ϕ (degree) | Unit Weight (pcf) |
|---|---|---|---|---|
| 1 – 2 | Very Loose | 0 – 15 | 26 – 28 | 70 – 100 |
| 3 – 6 | Loose | 15 – 35 | 28 – 30 | 90 – 115 |
| 7 – 15 | Medium | 35 – 65 | 30 – 34 | 110 – 130 |
| 16 – 30 | Dense | 65 – 85 | 33 – 38 | 110 – 140 |
| >30 | Very Dense | >85 | >50 | 130 – 150 |

**Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)**

| N values | Compactness | Relative Density, Dr % | ϕ (degree) | Unit Weight (pcf) |
|---|---|---|---|---|
| 2 – 3 | Very Loose | 0 – 15 | 27 – 28 | 70 – 100 |
| 4 – 7 | Loose | 15 – 35 | 30 – 32 | 90 – 115 |
| 8 – 20 | Medium | 35 – 65 | 32 – 36 | 110 – 130 |
| 21 – 40 | Dense | 65 – 85 | 36 – 42 | 110 – 140 |

**Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)**

| N values | Compactness | Relative Density, Dr % | ϕ (degree) | Unit Weight (pcf) |
|---|---|---|---|---|
| 3 – 6 | Very Loose | 0 – 15 | 28 – 30 | 70 – 100 |
| 5 – 9 | Loose | 15 – 35 | 30 – 34 | 90 – 115 |
| 10 – 25 | Medium | 35 – 65 | 33 – 40 | 110 – 130 |
| 26 – 40 | Dense | 65 – 85 | 40 – 50 | 110 – 140 |
| >40 | Very Dense | >85 | >50 | 130 – 150 |

**<u>Bearing capacity for deep foundation</u>**

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

**Table 6: Formula Used to Determine Bearing Capacity of Soil**

| **Shallow Foundation** | |
|---|---|
| For Cohesive Soil | |
| $q_a = q_u + \gamma \frac{D}{3}$<br>$q_a = q_u$(Common Practice) | Where, $q_a$=Allowable bearing capacity, $q_u$= Unconfined Compressive Strength, $\gamma$= Unit Weight of soil, $D$= Depth of footing |
| For Cohesion less Soil | |
| $q_{ult} = 30N$;<br>$q_a = q_{ult}/FS$ | $q_{ult}$= Ultimate bearing capacity in kPa, $N$ = SPT Number, $FS$= Factor of Safety |
| **Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition** | |
| **Deep Foundation** | |
| For Cohesive Soil | |
| $f_{su} = \alpha C_u$ ;<br>$q_{pu} = N_c C_u$ | $\alpha$ = Adhesion factor, $f_{su}$ = Ultimate skin friction, $C_u$ = Undrained Shear Strength, $q_{pu}$ = Ultimate end bearing of pile, $N_c$ = Bearing capacity factor for deep foundation=9 |
| For Cohesion less Soil | |
| $f_{su} = N/50\ tsf$ ;<br>$q_{pu} = 4N\ tsf$ | $f_{su}$ = Ultimate skin friction, $q_{pu}$ = Ultimate end bearing of pile, $N$ = SPT Number |
| **Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"** | |

Based on the above formula, the important parameters for foundation design have been determined and are given in **Table 7**, **Table 8, Table 9, Table 10 and Table 11** respectively.

**Table 7: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5)**

| **Bore Hole No.** | **Depth** | **Design N** | **Clay** | | | | | | **Sand** |
|---|---|---|---|---|---|---|---|---|---|
| | | | **Continuous** | | | **Square (Isolated)** | | | |
| | **(ft)** | **-** | **D/B=0.25** | **D/B=0.5** | **D/B=1** | **D/B=0.25** | **D/B=0.5** | **D/B=1** | |
| **BH01** | **2.5-5.0** | **11** | 1.253 | 1.318 | 1.436 | 1.480 | 1.598 | 1.739 | 1.350 |
| | **5.0-10** | **17** | 1.956 | 2.057 | 2.243 | 2.310 | 2.496 | 2.715 | 2.108 |
| **BH02** | **2.5-5.0** | **11** | 1.230 | 1.293 | 1.410 | 1.452 | 1.569 | 1.707 | 1.325 |
| | **5.0-10** | **14** | 1.680 | 1.767 | 1.926 | 1.984 | 2.143 | 2.332 | 1.810 |
| **BH03** | **2.5-5.0** | **13** | 1.485 | 1.562 | 1.702 | 1.754 | 1.894 | 2.061 | 1.600 |
| | **5.0-10** | **15** | 1.776 | 1.868 | 2.036 | 2.098 | 2.266 | 2.465 | 1.914 |
| **BH04** | **2.5-5.0** | **13** | 1.462 | 1.537 | 1.676 | 1.726 | 1.865 | 2.029 | 1.575 |
| | **5.0-10** | **11** | 1.244 | 1.308 | 1.426 | 1.469 | 1.587 | 1.727 | 1.341 |
| **BH05** | **2.5-5.0** | **7** | 0.766 | 0.805 | 0.878 | 0.904 | 0.977 | 1.063 | 0.825 |
| | **5.0-10** | **11** | 1.300 | 1.367 | 1.491 | 1.535 | 1.659 | 1.804 | 1.401 |
| **BH06** | **2.5-5.0** | **12** | 1.438 | 1.513 | 1.649 | 1.699 | 1.835 | 1.996 | 1.550 |
| | **5.0-10** | **16** | 1.860 | 1.956 | 2.133 | 2.197 | 2.373 | 2.582 | 2.004 |
| **BH07** | **2.5-5.0** | **0** | - | - | - | - | - | - | - |
| | **5.0-10** | **0** | - | - | - | - | - | - | - |
| **BH08** | **2.5-5.0** | **0** | - | - | - | - | - | - | - |
| | **5.0-10** | **0** | - | - | - | - | - | - | - |
| **BH09** | **2.5-5.0** | **0** | - | - | - | - | - | - | - |
| | **5.0-10** | **0** | - | - | - | - | - | - | - |
| **BH10** | **2.5-5.0** | **0** | - | - | - | - | - | - | - |
| | **5.0-10** | **0** | - | - | - | - | - | - | - |

**Table 8: The allowable skin friction for Precast driven piles in Tsf (FS=2.5)**

| Depth | Bore Hole | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ft | BH01 | BH02 | BH03 | BH04 | BH05 | BH06 | BH07 | BH08 | BH09 | BH10 |
| 5 | 0.049 | 0.074 | 0.049 | 0.074 | 0.074 | 0.049 | - | - | - | - |
| 10 | 0.168 | 0.160 | 0.128 | 0.152 | 0.118 | 0.200 | - | - | - | - |
| 15 | 0.118 | 0.118 | 0.120 | 0.239 | 0.218 | 0.118 | - | - | - | - |
| 20 | 0.200 | 0.152 | 0.120 | 0.074 | 0.144 | 0.152 | - | - | - | - |
| 25 | 0.168 | 0.096 | 0.118 | 0.096 | 0.138 | 0.118 | - | - | - | - |
| 30 | 0.138 | 0.156 | 0.168 | 0.218 | 0.218 | 0.138 | - | - | - | - |
| 35 | 0.138 | 0.174 | 0.096 | 0.118 | 0.118 | 0.096 | - | - | - | - |
| 40 | 0.156 | 0.196 | 0.074 | 0.138 | 0.118 | 0.118 | - | - | - | - |
| 45 | 0.196 | 0.218 | 0.174 | 0.156 | 0.138 | 0.096 | - | - | - | - |
| 50 | 0.200 | 0.200 | 0.200 | 0.240 | 0.334 | 0.321 | - | - | - | - |
| 55 | 0.192 | 0.192 | 0.669 | 0.535 | 0.562 | 0.428 | - | - | - | - |
| 60 | - | - | - | 0.535 | 0.669 | 0.669 | - | - | - | - |
| 65 | - | - | - | - | - | - | - | - | - | - |
| 70 | - | - | - | - | - | - | - | - | - | - |
| 75 | - | - | - | - | - | - | - | - | - | - |
| 80 | - | - | - | - | - | - | - | - | - | - |
| 85 | - | - | - | - | - | - | - | - | - | - |
| 90 | - | - | - | - | - | - | - | - | - | - |
| 95 | - | - | - | - | - | - | - | - | - | - |
| 100 | - | - | - | - | - | - | - | - | - | - |

**Table 9: The allowable end bearing for Precast driven piles in Tsf (FS=2.5)**

| Depth | Bore Hole | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ft | BH01 | BH02 | BH03 | BH04 | BH05 | BH06 | BH07 | BH08 | BH09 | BH10 |
| 5 | 0.450 | 0.675 | 0.450 | 0.675 | 0.675 | 0.450 | - | - | - | - |
| 10 | 33.600 | 32.000 | 25.600 | 30.400 | 1.125 | 40.000 | - | - | - | - |
| 15 | 1.125 | 1.125 | 24.000 | 2.475 | 2.250 | 1.125 | - | - | - | - |
| 20 | 40.000 | 30.400 | 24.000 | 0.675 | 28.800 | 30.400 | - | - | - | - |
| 25 | 33.600 | 0.900 | 1.125 | 0.900 | 1.350 | 1.125 | - | - | - | - |
| 30 | 1.350 | 1.575 | 33.600 | 2.250 | 2.250 | 1.350 | - | - | - | - |
| 35 | 1.350 | 1.800 | 0.900 | 1.125 | 1.125 | 0.900 | - | - | - | - |
| 40 | 1.575 | 2.025 | 0.675 | 1.350 | 1.125 | 1.125 | - | - | - | - |
| 45 | 2.025 | 2.250 | 1.800 | 1.575 | 1.350 | 0.900 | - | - | - | - |
| 50 | 40.000 | 40.000 | 40.000 | 3.825 | 5.625 | 5.400 | - | - | - | - |
| 55 | 38.400 | 38.400 | 11.250 | 9.000 | 9.450 | 7.200 | - | - | - | - |
| 60 | - | - | - | 9.000 | 11.250 | 11.250 | - | - | - | - |
| 65 | - | - | - | - | - | - | - | - | - | - |
| 70 | - | - | - | - | - | - | - | - | - | - |
| 75 | - | - | - | - | - | - | - | - | - | - |
| 80 | - | - | - | - | - | - | - | - | - | - |
| 85 | - | - | - | - | - | - | - | - | - | - |
| 90 | - | - | - | - | - | - | - | - | - | - |
| 95 | - | - | - | - | - | - | - | - | - | - |
| 100 | - | - | - | - | - | - | - | - | - | - |

**Table 10: The allowable skin friction for Bored piles in Tsf (FS=2.5)**

| Depth | Bore Hole | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ft | BH01 | BH02 | BH03 | BH04 | BH05 | BH06 | BH07 | BH08 | BH09 | BH10 |
| 5 | 0.032 | 0.048 | 0.032 | 0.048 | 0.048 | 0.032 | - | - | - | - |
| 10 | 0.109 | 0.104 | 0.083 | 0.099 | 0.077 | 0.130 | - | - | - | - |
| 15 | 0.077 | 0.077 | 0.078 | 0.156 | 0.141 | 0.077 | - | - | - | - |
| 20 | 0.130 | 0.099 | 0.078 | 0.048 | 0.094 | 0.099 | - | - | - | - |
| 25 | 0.109 | 0.062 | 0.077 | 0.062 | 0.090 | 0.077 | - | - | - | - |
| 30 | 0.090 | 0.101 | 0.109 | 0.141 | 0.141 | 0.090 | - | - | - | - |
| 35 | 0.090 | 0.113 | 0.062 | 0.077 | 0.077 | 0.062 | - | - | - | - |
| 40 | 0.101 | 0.127 | 0.048 | 0.090 | 0.077 | 0.077 | - | - | - | - |
| 45 | 0.127 | 0.141 | 0.113 | 0.101 | 0.090 | 0.062 | - | - | - | - |
| 50 | 0.130 | 0.130 | 0.130 | 0.156 | 0.217 | 0.209 | - | - | - | - |
| 55 | 0.125 | 0.125 | 0.435 | 0.348 | 0.365 | 0.278 | - | - | - | - |
| 60 | - | - | - | 0.348 | 0.435 | 0.435 | - | - | - | - |
| 65 | - | - | - | - | - | - | - | - | - | - |
| 70 | - | - | - | - | - | - | - | - | - | - |
| 75 | - | - | - | - | - | - | - | - | - | - |
| 80 | - | - | - | - | - | - | - | - | - | - |
| 85 | - | - | - | - | - | - | - | - | - | - |
| 90 | - | - | - | - | - | - | - | - | - | - |
| 95 | - | - | - | - | - | - | - | - | - | - |
| 100 | - | - | - | - | - | - | - | - | - | - |

**Table 11: The allowable end bearing for Bored piles in Tsf (FS=3.0)**

| Depth | Bore Hole | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ft | BH01 | BH02 | BH03 | BH04 | BH05 | BH06 | BH07 | BH08 | BH09 | BH10 |
| 5 | 0.203 | 0.304 | 0.203 | 0.304 | 0.304 | 0.203 | - | - | - | - |
| 10 | 15.120 | 14.400 | 11.520 | 13.680 | 0.506 | 18.000 | - | - | - | - |
| 15 | 0.506 | 0.506 | 10.800 | 1.114 | 1.013 | 0.506 | - | - | - | - |
| 20 | 18.000 | 13.680 | 10.800 | 0.304 | 12.960 | 13.680 | - | - | - | - |
| 25 | 15.120 | 0.405 | 0.506 | 0.405 | 0.608 | 0.506 | - | - | - | - |
| 30 | 0.608 | 0.709 | 15.120 | 1.013 | 1.013 | 0.608 | - | - | - | - |
| 35 | 0.608 | 0.810 | 0.405 | 0.506 | 0.506 | 0.405 | - | - | - | - |
| 40 | 0.709 | 0.911 | 0.304 | 0.608 | 0.506 | 0.506 | - | - | - | - |
| 45 | 0.911 | 1.013 | 0.810 | 0.709 | 0.608 | 0.405 | - | - | - | - |
| 50 | 18.000 | 18.000 | 18.000 | 1.721 | 2.531 | 2.430 | - | - | - | - |
| 55 | 17.280 | 17.280 | 5.063 | 4.050 | 4.253 | 3.240 | - | - | - | - |
| 60 | - | - | - | 4.050 | 5.063 | 5.063 | - | - | - | - |
| 65 | - | - | - | - | - | - | - | - | - | - |
| 70 | - | - | - | - | - | - | - | - | - | - |
| 75 | - | - | - | - | - | - | - | - | - | - |
| 80 | - | - | - | - | - | - | - | - | - | - |
| 85 | - | - | - | - | - | - | - | - | - | - |
| 90 | - | - | - | - | - | - | - | - | - | - |
| 95 | - | - | - | - | - | - | - | - | - | - |
| 100 | - | - | - | - | - | - | - | - | - | - |

# CONCLUSION AND RECOMMENDATION

On the basis of analysis and information provided by the owner following recommendations are suggested for the proposed six storied residential building.

***Soil profile scenario -*** Overall soil information of the investigated site is quite irregular in between the boreholes and remarkable vertical inconsistency exists in the soil formation. Four types of scenarios have been observed among the boreholes.

For borehole 01, 02, 03–Grey sandy clay and grey soft clay with low bearing capacity exists up to 5 ft depth from EGL. After that alternate layer of grey sandy clay and grey fine sand with low to moderate bearing capacity have been noticed. However, black organic soil layer with low bearing capacity has been observed from 25 ft to 35 ft depth in borehole 02 and 03 from EGL. Then a layer of grey soft clay with low bearing capacity initiates and continues up to 45 ft depth in each borehole. Grey fine sand and grey hard clay with good bearing capacity has been found from 50 ft to 55 ft depth from EGL as the client provided.

For borehole 04–Grey fine sand with moderate bearing capacity has been observed at 10 ft depth from EGL. After that, grey sandy clay with low bearing capacity exists at 15 ft depth from EGL. Grey soft clay with low bearing capacity has been found from 20 ft to 45 ft depth from EGL. A layer of grey stiff clay with moderate bearing capacity has been noticed at 50 ft depth from EGL. Then brown hard clay with good bearing capacity initiates at 50 ft depth and continues to 60 ft depth from EGL as the client provided.

For borehole 05–Grey sandy clay with low bearing capacity exists up to 15 ft depth from EGL. After that grey fine sand with moderate bearing capacity has been observed at 20 ft depth from EGL. Grey sandy clay with low bearing capacity has also been noticed from 25 ft to 45 ft depth from EGL. Brown stiff clay and brown hard clay with adequate bearing capacity have been seen from 50 ft to 60 ft depth from EGL as the client provided.

For borehole 06–Grey soft clay with low bearing capacity exists up to 5 ft depth from EGL. After that grey fine sand with moderate bearing capacity and grey soft clay with low bearing capacity have been noticed from 10 ft to 20 ft depth from EGL. Grey sandy clay with low bearing capacity has been observed from 25 ft to 45 ft depth from EGL. Brown stiff clay and brown hard clay with moderate to good bearing capacity have been found from 50 ft to 60 ft depth from EGL as the information provided by the client.

***Foundation options -*** In this situation, the foundation should be selected considering both the dimension and design load for the proposed building. Since soft clay soil with very low bearing capacity has been found in the top layers, shallow foundation should not be considered in this case. Deep foundation (bored/precast pile) may be selected in this case to ensure the stability of the structure. For designing any suitable deep foundation, the necessary data for skin friction and end bearing can be found from the attached **table (8 - 11)** in the report, which is based on existing soil condition.

***Safety provisions -*** It is strongly recommended to check the **settlement** of the site considering building load while designing either shallow or deep foundation. Obviously, **pile load test** covering the entire project area must be carried out for confirming the design parameters of the service pile (*if pile foundation is considered*). Moreover, the earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be considered for the safe design of shallow or deep foundation.

***Attachments -*** The layout of the boreholes of the investigated site and the bore-log profiles with SPT values are shown in the Appendix. Client performed no special laboratory test for the proposed site.

***Special notes -*** These suggestions are tentative, and selection of type, depth, and dimensions of foundation would be decided by the Consultant or Design Engineer. It is also to be noted that the above recommendation is restricted only to the existing soil condition of **55-60 ft** as client provided.

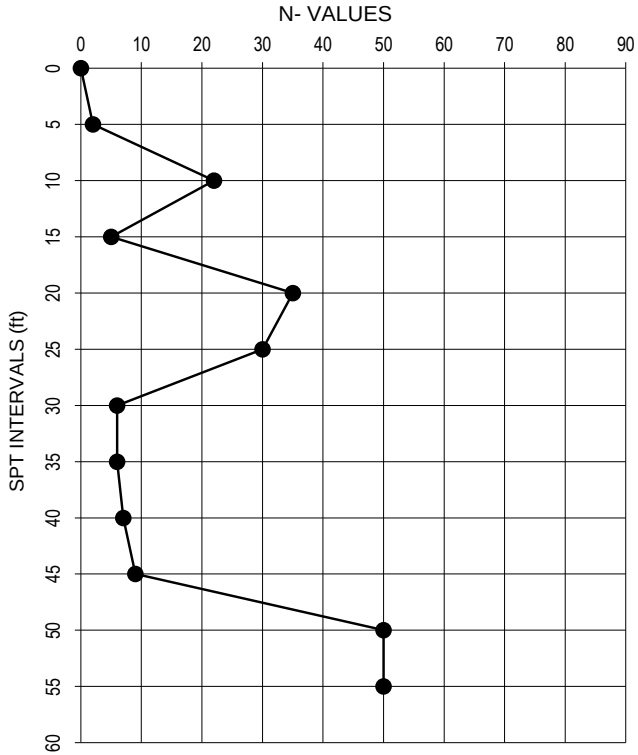
| **Test Assisted By** | **Report Prepared By** | **Countersigned By** |
|---|---|---|
| **Md. Nazrul Islam** | **Ayesha Ferdous Mita** | **Prof. Dr. H M A Mahzuz** |
| Senior Technical Officer<br>Dept. of CEE, SUST-3114 | Lecturer<br>Dept. of CEE, SUST-3114 | Coordinator, CRTC<br>Dept. of CEE, SUST-3114 |

# RECORD OF BORING AND TESTING

| | | | |
|---|---|---|---|
| CLIENT : **MOJNU MIAH** | LOCATION: Megna B-52, Dariapara, Sylhet. | | Sandy clay with silt. |
| PROJECT NAME : **CONST. OF 06 STORIED RESIDENTIAL BUILDING .** | GROUND WATER LEVEL : 3'-0" BELOW EGL. | | |
| BORE HOLE NO. **01** | | | Clay with silt & sand. |
| METHOD OF BORING : **WASH.** | STARTING DATE : 07/03/2022 | | |
| BORING DIA : **0.33ft** | | | Sand with silt & clay. |
| DEPTH OF BORING : 55 ft | COMPLETION DATE : 07/03/2022 | | |
| | | | Organic/Gravel/coarse sand |

| Sample No. | Type of sample | Depth Below GL(ft) | Thickness(ft) | DESCRIPTION OF SOIL STRATA | SYMBOL | SPT Intervals(ft) | 06 inch | 06 inch | 06 inch | N- Value |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 0 | 0 | 0 | 0 | 0 |
| D1 | | 5.0 | 5.0 | Grey sandy clay | | 5.0 | 1 | 1 | 1 | 2 |
| D2 | | 10.0 | 5.0 | Grey fine sand | | 10.0 | 7 | 10 | 12 | 22 |
| D3 | | 15.0 | 5.0 | Grey sandy clay | | 15.0 | 2 | 2 | 3 | 5 |
| D4 | | 20.0 | 10.0 | Grey fine sand | | 20.0 | 6 | 15 | 20 | 35 |
| D5 | | 25.0 | | | | 25.0 | 5 | 13 | 17 | 30 |
| D6 | | 30.0 | 20.0 | Grey soft clay | | 30.0 | 2 | 3 | 3 | 6 |
| D7 | | 35.0 | | | | 35.0 | 2 | 3 | 3 | 6 |
| D8 | | 40.0 | | | | 40.0 | 2 | 3 | 4 | 7 |
| D9 | | 45.0 | | | | 45.0 | 3 | 4 | 5 | 9 |
| D10 | | 50.0 | 10.0 | Grey fine sand | | 50.0 | 14 | 23 | 27 | 50 |
| D11 | | 55.0 | | | | 55.0 | 16 | 24 | 26 | 50 |

GRAPHICAL REPRESANTATION OF N- VALUES

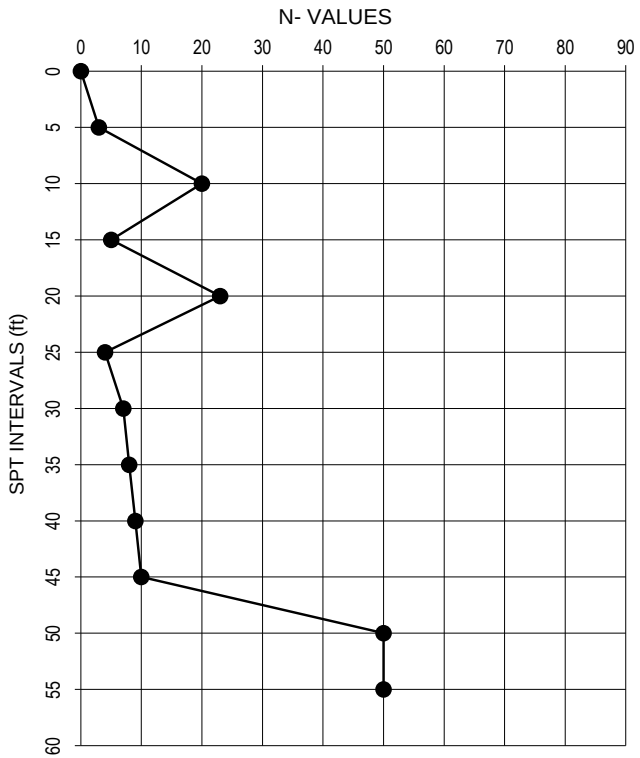


# RECORD OF BORING AND TESTING

| | | | |
|---|---|---|---|
| CLIENT : **MOJNU MIAH** | LOCATION: Megna B-52, Dariapara, Sylhet. | | Sandy clay with silt. |
| PROJECT NAME : **CONST. OF 06 STORIED RESIDENTIAL BUILDING .** | GROUND WATER LEVEL : 3'-0" BELOW EGL. | | |
| BORE HOLE NO. **02** | | | Clay with silt & sand. |
| METHOD OF BORING : **WASH.** | STARTING DATE : 07/03/2022 | | |
| BORING DIA : **0.33ft** | | | Sand with silt & clay. |
| DEPTH OF BORING : 55 ft | COMPLETION DATE : 07/03/2022 | | |
| | | | Organic/Gravel/coarse sand |

| Sample No. | Type of sample | Depth Below GL(ft) | Thickness(ft) | DESCRIPTION OF SOIL STRATA | SYMBOL | SPT Intervals(ft) | 06 inch | 06 inch | 06 inch | N- Value |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 0 | 0 | 0 | 0 | 0 |
| D1 | | 5.0 | 5.0 | Grey soft clay | | 5.0 | 1 | 1 | 2 | 3 |
| D2 | | 10.0 | 5.0 | Grey fine sand | | 10.0 | 6 | 8 | 12 | 20 |
| D3 | | 15.0 | 5.0 | Grey soft clay | | 15.0 | 2 | 2 | 3 | 5 |
| D4 | | 20.0 | 5.0 | Grey fine sand | | 20.0 | 6 | 10 | 13 | 23 |
| D5 | | 25.0 | 15.0 | Black Organic | | 25.0 | 2 | 2 | 2 | 4 |
| D6 | | 30.0 | | | | 30.0 | 2 | 3 | 4 | 7 |
| D7 | | 35.0 | | | | 35.0 | 2 | 3 | 5 | 8 |
| D8 | | 40.0 | 10.0 | Grey soft clay | | 40.0 | 3 | 4 | 5 | 9 |
| D9 | | 45.0 | | | | 45.0 | 4 | 4 | 6 | 10 |
| D10 | | 50.0 | 10.0 | Grey fine sand | | 50.0 | 14 | 22 | 28 | 50 |
| D11 | | 55.0 | | | | 55.0 | 16 | 23 | 27 | 50 |

GRAPHICAL REPRESANTATION OF N- VALUES

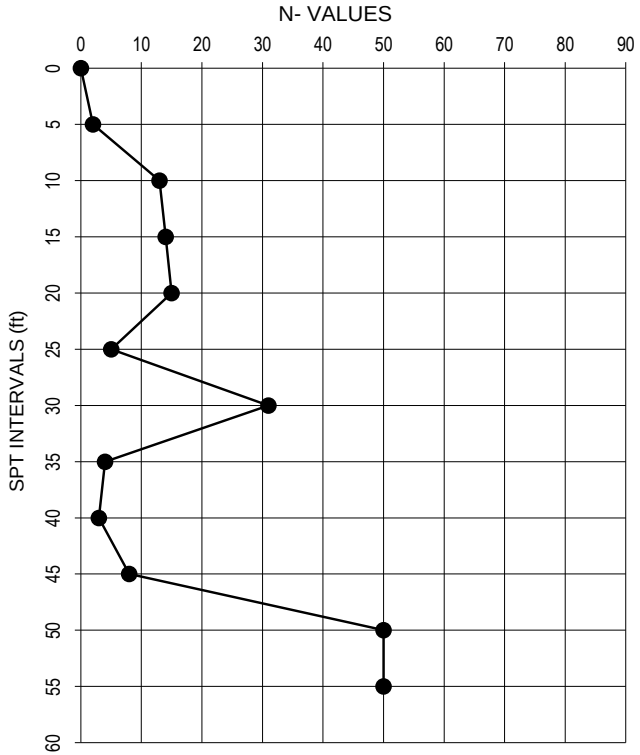


# RECORD OF BORING AND TESTING

| | | | |
|---|---|---|---|
| CLIENT : **MOJNU MIAH** | LOCATION: Megna B-52, Dariapara, Sylhet. | | Sandy clay with silt. |
| PROJECT NAME : **CONST. OF 06 STORIED RESIDENTIAL BUILDING .** | GROUND WATER LEVEL : 5'-0" BELOW EGL. | | |
| BORE HOLE NO. **03** | | | Clay with silt & sand. |
| METHOD OF BORING : **WASH.** | STARTING DATE : 07/03/2022 | | |
| BORING DIA : **0.33ft** | | | Sand with silt & clay. |
| DEPTH OF BORING : 55 ft | COMPLETION DATE : 07/03/2022 | | |
| | | | Organic/Gravel/coarse sand |

| Sample No. | Type of sample | Depth Below GL(ft) | Thickness(ft) | DESCRIPTION OF SOIL STRATA | SYMBOL | SPT Intervals(ft) | 06 inch | 06 inch | 06 inch | N- Value |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 0 | 0 | 0 | 0 | 0 |
| D1 | | 5.0 | 5.0 | Grey sandy clay | | 5.0 | 1 | 1 | 1 | 2 |
| D2 | | 10.0 | 15.0 | Grey fine sand | | 10.0 | 4 | 5 | 8 | 13 |
| D3 | | 15.0 | | | | 15.0 | 5 | 6 | 8 | 14 |
| D4 | | 20.0 | | | | 20.0 | 7 | 10 | 5 | 15 |
| D5 | | 25.0 | 5.0 | Black Organic | | 25.0 | 2 | 2 | 3 | 5 |
| D6 | | 30.0 | 5.0 | Grey fine sand | | 30.0 | 7 | 14 | 17 | 31 |
| D7 | | 35.0 | 5.0 | Black Organic | | 35.0 | 2 | 2 | 2 | 4 |
| D8 | | 40.0 | 10.0 | Grey soft clay | | 40.0 | 1 | 1 | 2 | 3 |
| D9 | | 45.0 | | | | 45.0 | 2 | 4 | 4 | 8 |
| D10 | | 50.0 | 5.0 | Grey fine sand | | 50.0 | 13 | 22 | 28 | 50 |
| D11 | | 55.0 | 5.0 | Grey hard clay | | 55.0 | 15 | 24 | 26 | 50 |

GRAPHICAL REPRESANTATION OF N- VALUES

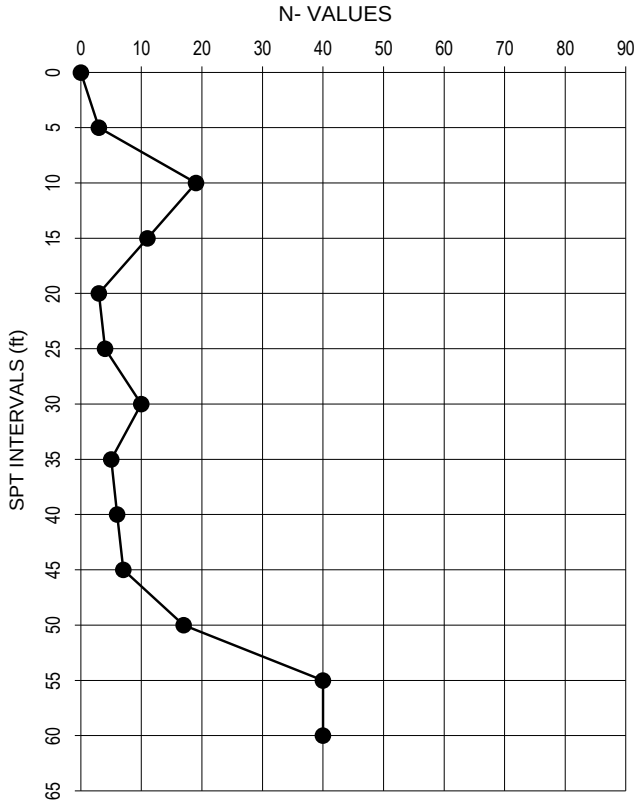


# RECORD OF BORING AND TESTING

| | | | |
|---|---|---|---|
| CLIENT : **MOJNU MIAH** | LOCATION: Megna B-52, Dariapara, Sylhet. | | Sandy clay with silt. |
| PROJECT NAME : **CONST. OF 06 STORIED RESIDENTIAL BUILDING .** | GROUND WATER LEVEL : 5'-2" BELOW EGL. | | |
| BORE HOLE NO. **04** | | | Clay with silt & sand. |
| METHOD OF BORING : **WASH.** | STARTING DATE : 08/03/2022 | | |
| BORING DIA : **0.33ft** | | | Sand with silt & clay. |
| DEPTH OF BORING : 60 ft | COMPLETION DATE : 08/03/2022 | | |
| | | | Organic/Gravel/coarse sand |

| Sample No. | Type of sample | Depth Below GL(ft) | Thickness(ft) | DESCRIPTION OF SOIL STRATA | SYMBOL | SPT Intervals(ft) | 06 inch | 06 inch | 06 inch | N- Value |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 0 | 0 | 0 | 0 | 0 |
| D1 | | 5.0 | 5.0 | Soil Miss | | 5.0 | 1 | 1 | 2 | 3 |
| D2 | | 10.0 | 5.0 | Grey fine sand | | 10.0 | 6 | 8 | 11 | 19 |
| D3 | | 15.0 | 5.0 | Grey sandy clay | | 15.0 | 4 | 5 | 6 | 11 |
| D4 | | 20.0 | 30.0 | Grey soft clay | | 20.0 | 1 | 1 | 2 | 3 |
| D5 | | 25.0 | | | | 25.0 | 1 | 2 | 2 | 4 |
| D6 | | 30.0 | | | | 30.0 | 2 | 3 | 7 | 10 |
| D7 | | 35.0 | | | | 35.0 | 2 | 2 | 3 | 5 |
| D8 | | 40.0 | | | | 40.0 | 2 | 3 | 3 | 6 |
| D9 | | 45.0 | | | | 45.0 | 2 | 3 | 4 | 7 |
| D10 | | 50.0 | 5.0 | Grey stiff clay | | 50.0 | 3 | 7 | 10 | 17 |
| D11 | | 55.0 | 10.0 | Brown hard clay | | 55.0 | 9 | 18 | 22 | 40 |
| D12 | | 60.0 | | | | 60.0 | 13 | 13 | 27 | 40 |

GRAPHICAL REPRESANTATION OF N- VALUES

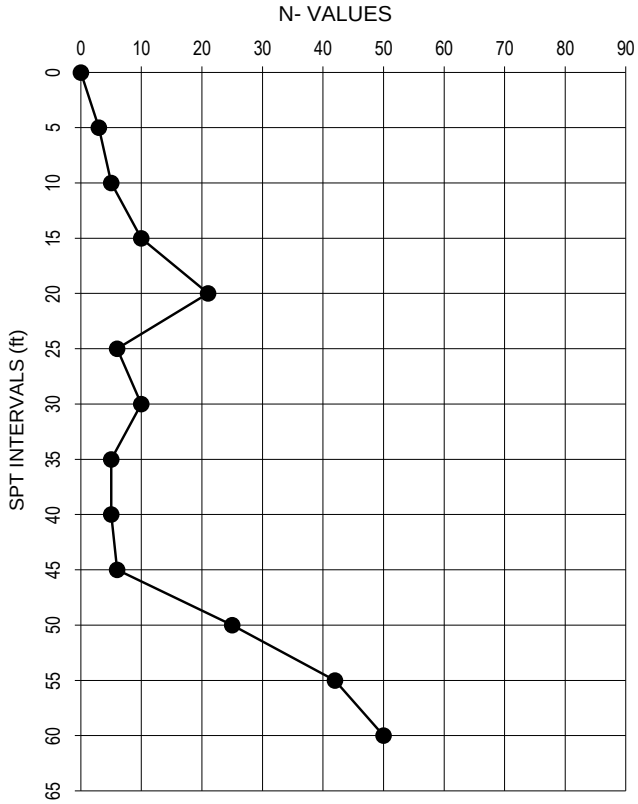


# RECORD OF BORING AND TESTING

| | | | |
|---|---|---|---|
| CLIENT : **MOJNU MIAH** | LOCATION: Megna B-52, Dariapara, Sylhet. | | Sandy clay with silt. |
| PROJECT NAME : **CONST. OF 06 STORIED RESIDENTIAL BUILDING .** | GROUND WATER LEVEL : 4'-0" BELOW EGL. | | |
| BORE HOLE NO. **05** | | | Clay with silt & sand. |
| METHOD OF BORING : **WASH.** | STARTING DATE : 08/03/2022 | | |
| BORING DIA : **0.33ft** | | | Sand with silt & clay. |
| DEPTH OF BORING : 60 ft | COMPLETION DATE : 08/03/2022 | | |
| | | | Organic/Gravel/coarse sand |

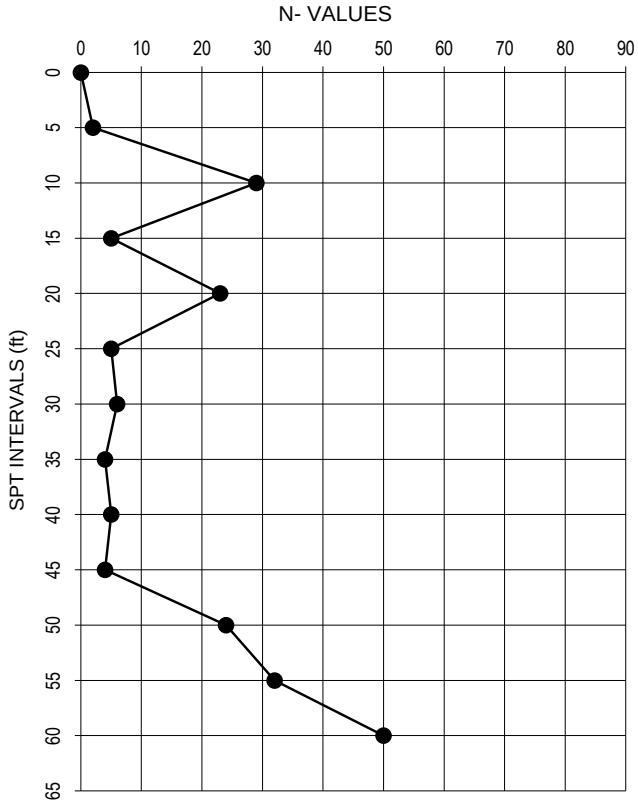
| Sample No. | Type of sample | Depth Below GL(ft) | Thickness(ft) | DESCRIPTION OF SOIL STRATA | SYMBOL | SPT Intervals(ft) | 06 inch | 06 inch | 06 inch | N- Value |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 0 | 0 | 0 | 0 | 0 |
| D1 | | 5.0 | 15.0 | Grey sandy clay | | 5.0 | 1 | 1 | 2 | 3 |
| D2 | | 10.0 | | | | 10.0 | 2 | 2 | 3 | 5 |
| D3 | | 15.0 | | | | 15.0 | 2 | 2 | 8 | 10 |
| D4 | | 20.0 | 5.0 | Grey fine sand | | 20.0 | 5 | 8 | 13 | 21 |
| D5 | | 25.0 | 25.0 | Grey sandy clay | | 25.0 | 2 | 3 | 3 | 6 |
| D6 | | 30.0 | | | | 30.0 | 3 | 5 | 5 | 10 |
| D7 | | 35.0 | | | | 35.0 | 2 | 2 | 3 | 5 |
| D8 | | 40.0 | | | | 40.0 | 2 | 2 | 3 | 5 |
| D9 | | 45.0 | | | | 45.0 | 2 | 3 | 3 | 6 |
| D10 | | 50.0 | 5.0 | Brown stiff clay | | 50.0 | 21 | 10 | 15 | 25 |
| D11 | | 55.0 | 10.0 | Brown hard clay | | 55.0 | 12 | 20 | 22 | 42 |
| D12 | | 60.0 | | | | 60.0 | 13 | 23 | 27 | 50 |

GRAPHICAL REPRESANTATION OF N- VALUES



# RECORD OF BORING AND TESTING

| | | | |
|---|---|---|---|
| CLIENT : **MOJNU MIAH** | LOCATION: Megna B-52, Dariapara, Sylhet. | | Sandy clay with silt. |
| PROJECT NAME : **CONST. OF 06 STORIED RESIDENTIAL BUILDING .** | GROUND WATER LEVEL : 2'-0" BELOW EGL. | | |
| BORE HOLE NO. **06** | | | Clay with silt & sand. |
| METHOD OF BORING : **WASH.** | STARTING DATE : 08/03/2022 | | |
| BORING DIA : **0.33ft** | | | Sand with silt & clay. |
| DEPTH OF BORING : 60 ft | COMPLETION DATE : 08/03/2022 | | |
| | | | Organic/Gravel/coarse sand |

| Sample No. | Type of sample | Depth Below GL(ft) | Thickness(ft) | DESCRIPTION OF SOIL STRATA | SYMBOL | SPT Intervals(ft) | 06 inch | 06 inch | 06 inch | N- Value |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 0 | 0 | 0 | 0 | 0 |
| D1 | | 5.0 | 5.0 | Grey soft clay | | 5.0 | 1 | 1 | 1 | 2 |
| D2 | | 10.0 | 5.0 | Grey fine sand | | 10.0 | 8 | 12 | 17 | 29 |
| D3 | | 15.0 | 5.0 | Grey soft clay | | 15.0 | 4 | 3 | 2 | 5 |
| D4 | | 20.0 | 5.0 | Grey fine sand | | 20.0 | 1 | 11 | 12 | 23 |
| D5 | | 25.0 | 25.0 | Grey sandy clay | | 25.0 | 2 | 2 | 3 | 5 |
| D6 | | 30.0 | | | | 30.0 | 3 | 3 | 3 | 6 |
| D7 | | 35.0 | | | | 35.0 | 2 | 2 | 2 | 4 |
| D8 | | 40.0 | | | | 40.0 | 2 | 2 | 3 | 5 |
| D9 | | 45.0 | | | | 45.0 | 1 | 1 | 3 | 4 |
| D10 | | 50.0 | 5.0 | Brown stiff clay | | 50.0 | 5 | 11 | 13 | 24 |
| D11 | | 55.0 | 10.0 | Brown hard clay | | 55.0 | 10 | 13 | 19 | 32 |
| D12 | | 60.0 | | | | 60.0 | 15 | 23 | 27 | 50 |

GRAPHICAL REPRESANTATION OF N- VALUES